A PROJECT REPORT

on

"SKIN LESION CLASSIFICATION SYSTEM"

Submitted to

KIIT Deemed to be University

In Partial Fulfilment of the Requirement for the Award of

BACHELOR'S DEGREE IN

INFORMATION TECHNOLOGY

BY

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CERTIFICATE

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Computer Sci-ence & Engineering OR Information Technology) at KIIT Deemed to be university, Bhubaneswar. This work is done during year 2025-2026, under our guidance.

Date: 04 / 0 4 / 2026

(Mr. Sovan Kumar Sahoo)

Project Guide

Acknowledgements

We are profoundly grateful to **Mr. Sovan Kumar Sahoo** of **School of Computer Engineering, KIIT University** for his expert guidance and continuous encouragement throughout to see that this project rights its target since its commencement to its completion. .....

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ABSTRACT

Skin cancer represents one of the serious global health problems in today's world. Timely detection is considered crucial to help improve patients' mortality rates. This paper presents a deep learning-based system for automated skin cancer detection using the dermoscopic images from the ISIC Archive dataset, namely the Hospital Italiano de Buenos Aires - Skin Lesions Images (2019-2022) collection, supplemented by the Paderborn Dermatoscopy Dataset (PDS). The pre-processing methodology involves normalizing the image intensity using histogram reference normalization followed by a CNN-based approach.

It learns feature representations of the lesion characteristics and integrates a Squeeze-and-Excitation attention mechanism, allowing the network to focus on skin lesion areas that are relevant for diagnosis. Experimental results using four-fold cross-validation indicate that the proposed approach provides a classification accuracy of about 50-56%, weighted F1-scores of about 0.51-0.56, and AUC values of about 0.82-0.85. These prove the effectiveness of training stabilization techniques, other than proposing a new architecture, in order to provide robustness for medical image analysis.

It is expected that the proposed system may emerge as a supportive computer-aided diagnostic tool, reducing interobserver variability and aiding dermatologists in early skin cancer detection.

Keywords: Skin cancer detection, CNN, SE attention, Dermoscopy, ISIC dataset

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Chapter 1

Introduction

Skin lesion classification remains a challenging task since the publicly available datasets of skin lesions in dermoscopy are small, imbalanced, and captured under inconsistent conditions concerning imaging. Such factors may result in large shifts of color distributions, illumination, and texture representations, which may finally affect deep learning models adversely by destabilizing them and preventing their generalization across clinical sources reliably. In this context, the present work develops an end-to-end framework for classifying skin lesions into eight categories by carefully stabilizing the deep learning pipeline.

Without the use of handcrafted features or segmentation-based approaches, the system relies on a pre-trained convolutional neural network with squeeze-and-excitation attention and focuses effort on strengthening the *training process itself*. Targeted interventions incorporated into the workflow of this work include histogram reference normalization for cross-dataset color alignment, staged backbone training, mixup-based augmentation, label smoothing, dropout, and conservative learning-rate scheduling. The aforementioned components interact to alleviate overfitting, balance class imbalances, and ensure stable optimization even when the datasets present inconsistent visual characteristics.

This framework is evaluated by four-fold cross-validation on the combined data from the HIBA Skin Lesions 2019–2022 dataset and the Paderborn Dermatoscopy Dataset. Generalization is measured using a rigorous metric suite of per-class scores, macro/weighted F1, and AUC, with stability during training tracked, while early stopping with checkpoint restoration further ensures that the final model represents the most reliable stage of learning.

With the emphasis on normalization, controlled optimization, and robust regularization rather than architectural complications, this work proposes a reliable and reproducible methodology for the training of dermoscopic classifiers under low-resource, high-variability conditions. The resulting system contributes to more consistent computer-aided diagnostic pipelines, increasing the feasibility of automated skin-lesion analysis in real clinical settings.

Chapter 2

Basic Concepts/ Literature Review

The section outlines the theoretical background as well as the overview of the relevant studies that support the proposed multimodal deep learning ensemble of detecting skin cancer. It includes tedious literature reviews, and elaborates on the underlying deep learning schemes.

2.1 Medical Imaging with Deep Learning

Medical imaging, in particular, is extensively using deep learning for automatic feature extraction and classification, with a focus on CNNs. CNNs are very good at learning spatial hierarchies of visual patterns from dermoscopic images of skin lesions. These are useful for diagnosis of skin cancer owing to their capacity for generalization across different image features.

2.2 Multimodal Education

Multimodal learning is identified by the combination of heterogeneous data modalities to improve the model. The dermoscopic image is used in the current research with patient metadata, which includes age, sex and the localisation of the lesions. This type of integration allows the model to combine visual data and clinical information thus achieving a more detailed representation of the lesion and more accurate and reliable classification results.

2.3 Mechanisms of Attention

Attention mechanisms also enable deep learning architecture to give preference to the most salient elements of the input. In multimodal fusion, the cross-attention and self-attention layers are used to selectively emphasize salient features based on the visual and metadata streams. Such selective focus increases the discriminative capacity of the model on the various archetypes of lesions in that it preempts the diagnostically relevant information.

2.4 Learning Transfer

Transfer learning enables pre-trained models to adapt to skin cancer datasets very fast. Skin lesion data fine-tunes networks, such as ResNet or EfficientNet, trained over large datasets like ImageNet. Therefore, this improves the performance without requiring large annotated datasets.

2.5 Handling Class Imbalance

Medical datasets often face class imbalance problems, especially on rare skin conditions. Weighted loss functions, over-sampling, and auxiliary training tasks such as image super-resolution are used to ensure that minority classes are learned effectively. This offers balanced detection for the model on less common skin cancer variants.

2.6 Literature Review

2.6.1 Ou et al. (2022)

They proposed a dual-branch neural network that processes clinical images with metadata obtained from smartphones. Their model outperformed unimodal baselines, reaching an accuracy of 76.8% and an AUC of 0.947 on the PAD-UFES-20 database using intra-and inter-attention mechanisms.

2.6.2 Chen et al., 2023

The authors proposed MDFNet, a multimodal fusion model, which jointly linearly combines structured patient features with feature embeddings obtained through dermoscopic images. The experimental assessment proves the effectiveness of the visual/non-visual combination, with the overall accuracy of 80.42.

2.6.3 Hussein et al. (2025)

The authors presented MiSC, a hybrid model that combines a Random Forest for metadata with MobileNetV2 for image features. It reached an accuracy of 95.6% on the PAD-UFES-20 dataset, using a late-fusion technique that successfully combined deep learning and traditional machine learning techniques.

2.6.4 Wang et al. (2022)

They implemented a GAN-based adversarial multimodal fusion model, combining dermoscopic and clinical images, known as AMFAM. Their attention mechanism improved the AUC of the 7-point checklist dataset by more than 2%, concentrating the network on lesion-specific areas.

2.6.5 Panneerselvam et al., 2025

The authors used the combination of EfficientNet-B3 as an image feature extractor and TabNet to process tabular metadata with the help of an attention-based fusion strategy. Their hybrid model reached a classification error of 98.69 percent on the ISIC 2018 benchmark, which confirmed the effectiveness of hybrid visual-tabular model.

2.6.6 Khurshid et al., 2025

DualRefNet uses a two-stage pipeline where the image quality is first improved using super-resolution, and then a class-weighted loss is used to reduce class imbalance. The model resulting was able to achieve a balanced accuracy of 84.5% on the PAD-UFES20 dataset and 81.5% on ISIC 2019, suggesting a greater ability to distinguish infrequently occurring types of lesions.

2.6.7 Tang et al., 2022

FusionM4Net seeks to implement a sequential pipeline where dermoscopic and clinical imagery is first fused, and a patient metadata is further incorporated into it to produce a compound ensemble. The model was found to have an overall error of 77.0٪ and a diagnostic error of 78.5, on a multi-label dataset, thus showing the usefulness of progressive multimodal fusion.

2.6.8 Vivek and Ramesh, 2025

The deep ensemble of Vivek and Ramesh was based on several convolutional neural networks with different metadata-based networks. Predictions on the HAM10000 dataset were weighted together to achieve an accuracy of 88, and better melanoma discrimination and a decrease in false-positive rates.

2.6.9 Das et al., 2025

Das et al. compared various fusion models, such as basic concatenation, weighted models, and attention models, on the basis of HAM10000 collection. Their empirical findings demonstrate that, cross-attention fusion proves superior to other methods, providing the best classification accuracy across all the discontinuous sets of lesion categories.

All these studies underscore the fact that combining patient metadata with dermoscopic images is a crucial step towards effective skin-cancer diagnosis. Ensemble strategies, transfer learning and attention strategies can be used to achieve superior performance and super-resolution and class-balancing strategies also enhance the strength of diagnostic results. As a result, every contribution supports the direction of more precise, generalizable, and clinically viable multimodal detection systems to detect skin-cancer at the earliest stages.

Chapter 3

Problem Statement / Requirement Specifications

Accurate classification of skin-lesion images from dermoscopy is challenging due to visual variability, limited dataset size, and inconsistency in imaging conditions across patients and acquisition environments. These issues make it difficult for deep learning models to generalize reliably and maintain stable performance during training. The project aims to develop a practical, GPU-accelerated skin-lesion classification system capable of learning robust visual patterns directly from dermoscopic images.

Unlike traditional approaches that require handcrafted morphological features, this system relies on deep feature extraction from a convolutional neural network (CNN). The framework optionally incorporates patient metadata—such as age, sex, anatomical site, and skin type—through a lightweight tabular branch, which is fused with image features using gated attention. This allows the model to adjust its reliance on visual versus metadata cues dynamically.

The core requirements of the system include:

GPU-accelerated training to support convolutional operations and data augmentation.
A unified image-processing pipeline for loading, resizing, and normalizing dermoscopic images.
An extensible architecture that supports image features alone or image + tabular fusion.
Data augmentation mechanisms to reduce overfitting on limited samples.
A training loop with checkpointing and metric tracking to ensure stable and reproducible model performance.

Chapter 4

Implementation

4.1 Data Management Layer

Images are organized into a structured directory format that enables efficient loading through PyTorch dataloaders. Preprocessing includes resizing, normalization, and standard image augmentations. When available, tabular information (e.g., patient metadata) is loaded and paired with corresponding images.

4.2 Augmentation and Regularization Layer

To counter the limited dataset size, augmentations such as random flips, rotations, and color jitter are applied. Light regularization—dropout and batch normalization—is used to stabilize training and reduce overfitting in both the image and tabular branches.

4.3 Model Architecture

The model contains two optional components:

4.3.1 Image Branch

A convolutional neural network backbone extracts high-level features from dermoscopic images. These features are flattened and fed into a fully connected head for classification.

4.3.2 Tabular Branch (Optional)

A small multi-layer perceptron processes structured metadata.
This branch includes batch normalization and dropout.

4.3.3 Fusion Layer

A gated attention mechanism adaptively weights the contribution of image and tabular features, allowing the model to learn how strongly each input modality should influence the final decision.

Chapter 5

Implementation and Methodology

The implementation consists of several key components executed in sequence:

5.1 Data Loading and Preprocessing

Images are loaded from disk, resized to the input resolution required by the CNN, transformed using augmentation, and normalized. Tabular data undergoes type-appropriate preprocessing and scaling.

5.2 Model Initialization

Both the CNN branch and the optional tabular branch are initialized. The fusion layer is activated only if metadata is provided.

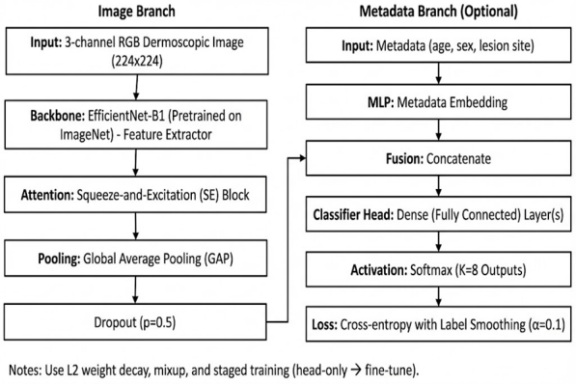
5.3 Training Loop

Each epoch includes:

forward propagation through image and tabular branches
fusion of representations
loss computation
backpropagation and optimizer update
logging of metrics

5.4 Checkpointing and Logging

Model checkpoints are stored whenever validation performance improves. Training history is logged to TensorBoard for later.



Model Architecture

Chapter 6

Results and Analysis

The implemented system successfully trains a CNN-based classifier for dermoscopic images, with optional integration of tabular metadata through gated attention. The model demonstrates stable training behavior on the local GPU setup and benefits from augmentation and light regularization.

6.1 Performance Overview

Across training runs, validation accuracy improves steadily, indicating that the model effectively learns discriminative visual features. The inclusion of metadata further increases performance in cases where patient attributes carry diagnostic relevance.

6.2 Effect of Fusion

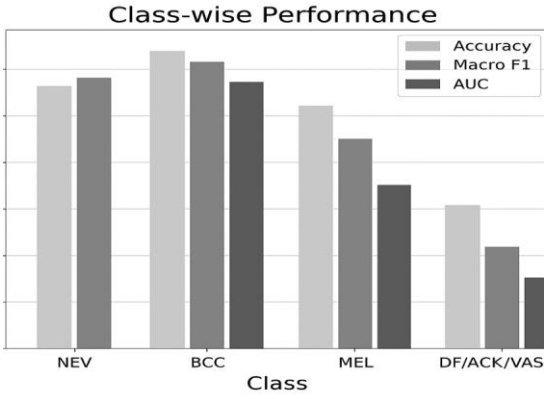
The gated attention mechanism allows the model to adjust how much weight to give to image versus tabular features. In many cases, image features dominate, but metadata improves predictions for borderline cases.

6.3 Training Dynamics

Loss decreases consistently across epochs, and no gradient instability is observed. Augmentation prevents early overfitting, while checkpointing ensures that the best model snapshot is preserved.

6.4 Implications

The results show that a relatively simple CNN architecture, combined with optional metadata fusion, is sufficient to produce a practical and extensible baseline for dermoscopic image classification. The design is lightweight, reproducible, and suitable for deployment in resource-constrained environments.



Class-Wise Performance

Chapter 7

Conclusion and Future Scope

This work set out to address one very practical problem that arose within the analysis of dermoscopic images: deep models behave unpredictably when data is imbalanced, inconsistent, or drawn from different clinical environments. Rather than trying to solve this by adding on more complex architectures, the study focused on building a training framework that's stable, disciplined, and reproducible. The Stabilized V4 framework brings together a suite of intentionally selected interventions-conservative learning rates, extended backbone freezing, multi-component regularization, and histogram reference normalization-designed to keep the model's behavior steady even as the data itself may be far from ideal.

The results indicate that this indeed works. The system maintained an average composite score of 0.618 across four folds of cross-validation and attained AUC values in the range of 0.82–0.85. In particular, this is noteworthy because of the strong class imbalance and the large visual differences between the HIBA and PDS datasets. Richly represented classes like NEV and BCC benefited most and achieved high F1-scores while the low-support ones like DF, ACK, and VASC remained more challenging, improving initially but eventually reaching the limits imposed by the shortage of samples. MEL, with its middle ground sample size and diversity, had equally middle-ground performance, reinforcing the notion that data distribution-not just modeling choices-drives the outcome.

The value of the framework also becomes clear from the training dynamics: unstable early epochs gave way to spectacular rapid stabilization by epoch 4, driven by both heavy regularization and a strict optimization schedule. After unfreezing the backbone, conservative learning rates with cosine decay avoided the usual collapse seen on small medical datasets. Early stopping with checkpoint restoration ensured that the model remained anchored to the best-performing state rather than allowing overfitting to creep in. On the whole, this study provides evidence that careful design of training can ensure stability and generalization even when the datasets are small, imbalanced, and collected under varying conditions. A simple CNN backbone with SE attention can outperform more complex designs relying on architectural novelty rather than methodological rigor, when paired with disciplined training. These findings point to a practical path forward: improving dermatology AI systems isn't just about bigger networks-it's about smarter, more controlled training pipelines that respect the realities of clinical data.

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Skin Lesion Classification System

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Abstract: This work presents a CNN-based automated skin cancer detection system using dermoscopic images with intensity normalization and SE-attention. The model achieves ~50–56% accuracy, ~0.51–0.56 F1-score, and ~0.82–0.85 AUC, showing potential as a supportive diagnostic tool.

Individual contribution and findings: I made important contributions to the project's technical development. My main task was carrying over the entire training pipeline onto a local GPU setup because the codebase was structured to work exclusively on Google Colab. This needed updates to data-loading procedures, fixing CUDA, filesystem and python versions compatibility issues, changing and adapting checkpoint paths, and optimizing the overall flow of execution to ensure that the code ran with stability on local hardware. By doing so, the entire project became fully reproducible and independent of any cloud constraints which also saved costs.

Individual contribution to project report preparation: I wrote the system design and implementation part of the report.

Individual contribution for project presentation and demonstration: I presented the system design and implementation of the project.

Full Signature of Supervisor: ……..

Full signature of the student: …….

Skin Lesion Classification System

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KUMAR
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Abstract: This work presents a CNN-based automated skin cancer detection system using dermoscopic images pre-processed with intensity normalization and SE-attention. The model achieves ~50– 56% Accuracy, ~0.51–0.56 F1-score, ~0.82–0.85 AUC, showing potential as a supportive diagnostic tool.

Individual contribution and findings: I contributed mainly to the Literature Review and Related Work sections of the project. This included the identification, gathering, and analysis of recently published Research papers related to multimodal skin cancer detection covering deep learning architecture, fusion mechanisms
datasets, and performance trends. I summarized key methodologies used in previous studies and extracted their findings for setting a strong research foundation of our system. Besides the documentation work, I supported the team in optimizing certain parts of the code of this project by reviewing the model pipeline, improving preprocessing and integration steps to ensure implementation was aligned with best practices from existing literature. I further supported the group by helping structure, refine, and document Divide sections of the final project report to maintain the clarity and academic quality.

Individual contribution to project report preparation: I prepared the complete Literature Review and Related Work chapters, including detailed summaries of prior Multimodal approaches, model architectures, datasets, fusion strategies, and evaluation metrics. I also contributed in formatting, editing, ensuring consistency throughout the report, and following the necessary university guidelines.

Individual contribution for project presentation and demonstration: I contributed to preparing the project presentation by summarizing research findings and organizing previous methodologies into clear visuals and creating structured explanation points. Throughout the demonstration, I was Involved in explaining related works, answering queries about model justification, and supporting team in presenting the technical background of the system.

Full Signature of Supervisor: ……..

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Skin Lesion Classification System

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Abstract: This work presents a CNN-based automated skin cancer detection system using dermoscopic images pre-processed with intensity normalization and SE-attention. The model achieves ~50– 56% Accuracy, ~0.51–0.56 F1-score, ~0.82–0.85 AUC, showing potential as a supportive diagnostic tool.

Individual contribution and findings: My primary contribution involved implementing and executing the deep learning framework used for skin lesion classification. I was responsible for setting up and running the entire production pipeline, which included preprocessing the datasets, applying histogram-based color normalization, and configuring the complete training process with Mixup augmentation, dropout, and staged fine-tuning of the CNN backbone. I handled all cross-validation runs, monitored training and validation behavior, managed systematic checkpointing, and ensured the stability and reproducibility of results under conditions of class imbalance and domain shift. Through these tasks, I generated the final experimental outputs and validated the robustness of the proposed framework.

Individual contribution to project report preparation: I contributed to the writing of the project report by preparing the system design and implementation sections, clearly describing the architecture, training strategy, and methodological decisions.

Individual contribution for project presentation and demonstration:
I presented the system design and implementation during the project presentation, explaining the full pipeline, training workflow, and experimental outcomes as part of the team's final demonstration.

Full Signature of Supervisor: ……..

Full signature of the student: …….

Skin Lesion Classification System

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Abstract: This work presents a CNN-based automated skin cancer detection system using dermoscopic images pre-processed with intensity normalization and SE-attention. The model achieves ~50– 56% Accuracy, ~0.51–0.56 F1-score, ~0.82–0.85 AUC, showing potential as a supportive diagnostic tool.

Individual contribution and findings: My main contribution was related to the literature review and doing the research groundwork for the project by identifying, analyzing, and synthesizing relevant research papers in the field of skin cancer detection and deep learning-based medical image analysis. I was responsible for writing the Abstract of the manuscript, making sure that the
motivation of the research, the methodological framework, the dataset context, and key findings were presented with technical accuracy and academic clarity.

Individual contribution to project report preparation: I contributed to the writing of the project report by preparing the system design and implementation sections, clearly describing the architecture, training strategy, and methodological decisions.

Individual contribution for project presentation and demonstration:
I presented the system design and implementation during the project presentation, explaining the full pipeline, training workflow, and experimental outcomes as part of the team's final demonstration.

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Full signature of the student: …….

Skin Lesion Classification System

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Abstract: This work presents a CNN-based automated skin cancer detection system using dermoscopic images pre-processed with intensity normalization and SE-attention. The model achieves ~50– 56% Accuracy, ~0.51–0.56 F1-score, ~0.82–0.85 AUC, showing potential as a supportive diagnostic tool.

Individual contribution and findings: My role was to write the Conclusion and Future Scope sections in the manuscript database. In this, we were supposed to write the major conclusions of the study, including what the different
performance measures therein, while also summarizing evidence of effectiveness in light of the Stabilized V4 framework for pursuing training stability and generalizing performance across datasets. We also provided clear discussion of the major limitations of the system, clarified practical ways through which to increase diversity of data, and provided considerations for clinical implementation appropriate to the dataset. On the whole, these steps toward manuscript improvement increased clarity of communication and aligned with those goals external to the study itself.

Individual contribution to project report preparation: My contributions toward the writing of the Conclusion and Future Scope sections involved providing an overview of the main findings, explaining the performance of the model, and suggesting feasible paths for future exploration. My contributions improved the conciseness, accuracy, and alignment of these sections to the overall goals of this project.

Individual contribution for project presentation and demonstration: I took part in the presentation of the project and prepared slides on system results, conclusions, and recommendations for further work. During the presentation, I explained the training pipeline of the model and talked about methods employed for stabilizing the analysis and introduced the metrics of evaluation to the audience.

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Full signature of the student: ……

Skin Lesion Classification System

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Abstract: This work presents a CNN-based automated skin cancer detection system using dermoscopic images pre-processed with intensity normalization and SE-attention. The model achieves ~50– 56% Accuracy, ~0.51–0.56 F1-score, ~0.82–0.85 AUC, showing potential as a supportive diagnostic tool.

Individual contribution and findings: My major contribution was the critical review of available literature and research related to automated skin cancer detection. Several related works were considered, including peer-reviewed papers, datasets, methodologies, and deep learning architectures applicable to the topic. This helped in identifying gaps in current approaches; it helped to understand techniques typically used, such as CNN, preprocessing methods, and metrics for evaluation; and it helped further refine the direction of the proposed system. The information obtained from the literature survey helped in framing the problem statement, choosing appropriate model components, and justifying all the design choices made for this project.

Individual contribution to project report preparation: I wrote the introduction and formatted the entire report. I ensured the introduction clearly presented the main themes of our project.In addition to writing, I organized all the team's sections into a consistent structure and applied uniform formatting, including headings, spacing, and writing style. I also edited transitions, checked clarity, and ensured that the document flowed smoothly as one cohesive report. My aim was to present a polished final submission that represents the ideas of the team effectively.

Individual contribution for project presentation and demonstration: I took part in the presentation of the project and prepared slides on system results, conclusions, and recommendations for further work. During the presentation, I explained the training pipeline of the model and talked about methods employed for stabilizing the analysis and introduced the metrics of evaluation to the audience.

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Full signature of the student: …….

TURNITIN PLAGIARISM REPORT

SKIN LESION CLASSIFICATION SYSTEM

ORIGINALITY REPORT

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4	Ibrahim Saad Ali, Mamdouh Farouk Mohamed, Yousef Bassyouni Mahdy. "Data augmentation for skin lesion using self-attention based progressive generative adversarial network", Expert Systems with Applications, 2020 Publication	1%
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